A Review of Polymer Enhanced Ultrafiltration Using Carboxymethyl Cellulose for Arsenic Removal

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ABSTRACT

Arsenic contamination in water poses a serious global threat due to its persistence, bioaccumulation, and severe health impacts. Conventional treatment techniques, such as coagulation, precipitation, and adsorption, often face limitations in achieving ultra-low discharge limits and may produce secondary waste. Polymer-enhanced ultrafiltration (PEUF) has emerged as a promising alternative that utilises a water-soluble polymer to selectively bind arsenic species before separation through ultrafiltration membranes. Carboxymethyl cellulose (CMC), a biodegradable and environmentally friendly polymer, has attracted significant attention due to its high density of carboxyl functional groups, which allow interactions with arsenic species under controlled pH conditions. Recent findings indicate that CMC-PEUF systems can achieve removal efficiencies of up to 94–98% for As(V) and 82–99% for As(III), depending on solution chemistry, polymer dosage, and membrane characteristics. The mechanisms of arsenic removal by CMC involve hydrogen bonding, localised electrostatic interactions, and cation-bridging effects rather than strong covalent bonding, enabling the formation of macromolecular complexes large enough to be retained by ultrafiltration membranes. Despite these advantages, challenges persist related to polymer recovery, membrane fouling, and selectivity in complex water matrices. This review summarises the principles of PEUF, the binding behaviour of CMC toward arsenic species, and the operational challenges associated with the application of CMC-PEUF in wastewater treatment systems. Furthermore, future directions are discussed to improve the scalability, selectivity, and economic feasibility of CMC-based PEUF for sustainable arsenic removal in wastewater treatment.

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Keywords: *arsenic; carboxymethyl cellulose; biosorbent, ultrafiltration; PEUF*

HIGHLIGHTS

- Carboxymethyl cellulose (CMC)- enhanced PEUF achieves high selectivity for arsenic removal in water bodies.
- CMC as biodegradable offers sustainable alternative to synthetic polymers.
- PEUF combines polymer complexation with ultrafiltration for efficient arsenic removal.
- CMC-PEUF maintains high water flux while minimizing membrane fouling.
- CMC-PEUF shows a strong potential for real-world wastewater treatment applications.

1 INTRODUCTION

Arsenic contamination in natural water bodies, particularly in wastewater sources, has become a major concern worldwide due to its serious implications on human health and environmental stability. Numerous studies have reported chronic exposure leading to severe diseases, including various cancers, cardiovascular complications, and neurological impairments (Khan, Gupta, Shrivastava, & Singh, 2022; L. Kumar et al., 2022; Mehdizadeh et al., 2022; Mojiri et al., 2024; Neumann et al., 2010). In Malaysia, alarming arsenic levels have been detected in several major water systems, including the Klang River Basin and

Kuantan River, often exceeding the recommended limits set by the World Health Organization (WHO) and the Malaysian Department of Environment (DOE). This has highlighted the urgent need to implement advanced treatment strategies to ensure the safety of drinking water and protect public health (Ahmed, Mokhtar, & Alam, 2021; Hubadillah et al., 2020; Jesi et al., 2024). The persistent nature of arsenic contamination, combined with its complex aqueous chemistry, particularly the highly toxic and mobile arsenate (As(III)) form, renders conventional water treatment technologies, such as coagulation-flocculation, adsorption, and basic membrane filtration inadequate for consistently achieving the ultra-low regulatory limits required, especially at trace level (Kabay et al., 2010; Rajendran, Garg, & Bajpai, 2021; Sathyamoorthy & Sneha, 2020). As a result, there is a need for hybrid water treatment processes that offer enhanced selectivity, operational efficiency, cost-effectiveness, and environmental compatibility, thereby providing a comprehensive solution to the shortcomings of conventional treatment techniques.

In this context, polymer-enhanced ultrafiltration (PEUF) has emerged as an innovative hybrid separation method that combines the advantages of chemical complexation and physical membrane separation into a single, highly effective process, allowing for the selective removal of specific ionic contaminants such as arsenic and minimal secondary pollution (Pookrod, Haller, & Scamehorn, 2005; Shen et al., 2024; Temiz & Arar, 2022). The principle of PEUF involves introducing water-soluble polymers containing functional groups capable of binding specific metal ions or other contaminants into the feed water, followed by the separation of the resulting macromolecular complexes through an ultrafiltration membrane, while allowing smaller unbound molecules to pass through (Huang & Feng, 2019; Korus, 2023). This dual mechanism significantly enhances the removal efficiency of specific contaminants such as arsenic, especially in complex matrices where other conventional methods might struggle due to interference from competing ions or organic matter present in wastewater (Bhattacharjee, Dutta, & Bhattacharjee, 2016; Qiu et al., 2023; Song et al., 2024). The operational parameters of PEUF, such as pH, polymer dosage, and membrane material selection, are crucial for the optimisation of binding and retention performance, making it a highly tunable and adaptable technology for different water treatment scenarios (Bin Darwish & AlAlawi, 2024; Fateev et al., 2023; Kraiem et al., 2024; Tapiero, Aravena, Vargas, Sánchez, & Huiliñir, 2025).

Carboxymethyl cellulose (CMC) has attracted significant attention as a polymer candidate for PEUF processes due to its unique combination of desirable properties, including its natural origin, biodegradability, high hydrophilicity. Additionally, the presence of abundant carboxyl functional groups makes it highly effective in binding to both positively or negatively charged metal species such as arsenic (Ab Rasid, Zainol, & Amin, 2021; Al-Mutair, Zubair, Kumar, Al-Mur, & Barakat, 2025; Rahman et al., 2021). The widespread availability and low cost of CMC further enhance its attractiveness as a sustainable and eco-friendly alternative to synthetic polymers. This aligns well with global environmental goals that promote the use of green materials in water purification technologies (Hussein, Hameed, & Jawad, 2022; Rahman et al., 2021). The inherent structural characteristics of CMC enable the formation of stable complexes with arsenic species across a wide pH range, particularly in mildly acidic to neutral conditions. These characteristics facilitate efficient separation via ultrafiltration membranes while maintaining high permeate flux and minimising the risk of membrane fouling (Zuriaga-Agustí et al., 2014).

This review article aims to critically examine the fundamental concepts underlying PEUF, with a particular focus on the use of carboxymethyl cellulose as a complexing agent for arsenic ion removal from aqueous systems. In addition, it discusses recent advancements, experimental findings, key operational challenges, and future perspectives related to the large-scale application of CMC-enhanced PEUF systems in real-world water treatment facilities, ultimately contributing to the broader goal of achieving safer, cleaner, and more sustainable water resources worldwide.

2 ARSENIC CONTAMINATION IN WASTEWATER

Arsenic, a naturally occurring metalloid, is ubiquitously present in the Earth's crust. As shown in Figure 1, its release into aquatic environments can occur through natural geochemical processes, such as mineral dissolution, volcanic emissions, and weathering of arsenic-bearing rocks, in addition to anthropogenic activities, including mining, the use of arsenical pesticides, wood preservation, and industrial effluent discharge (Hamid et al., 2024; Tabelin et al., 2018). Once released into water bodies, arsenic can exist in several chemical forms, primarily as arsenite (As(III)) and arsenate (As(V)), whose prevalence depends strongly on the prevailing redox potential and pH conditions of the environment. Under reducing conditions, As (III) dominates and poses a greater risk due to its higher mobility and toxicity. In contrast, under oxidizing conditions, arsenate is more stable and less harmful; nonetheless, it still requires careful monitoring and removal from wastewater system (Bagchi, Swaroop, & Bagchi, 2015; Pardeep Singh et al., 2021). The colorless, tasteless, and odorless nature of dissolved arsenic compounds renders their detection difficult without sophisticated analytical methods, thereby heightening the risk of inadvertent exposure in affected populations (Pardeep Singh et al., 2021).

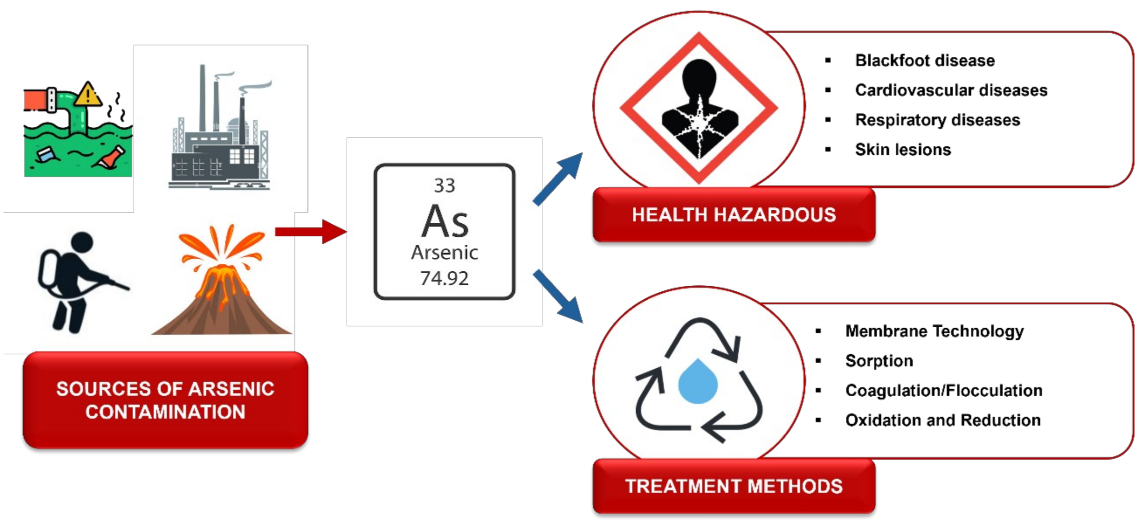

Figure 1: Schematic diagram of arsenic sources

Serious pollution of arsenic in the Malaysia's capital region has also been reported in recent years. High concentrations of arsenic (34.5 µg/L) were detected in surface water collected at the estuary of the Klang River basin (Ahmed et al., 2021). The arsenic concentration exceeded both the recommended limits and the raw water quality standard (10 µg/L) set by the Ministry of Health Malaysia (2009)(Ahmed et al., 2021). This underscores the urgency to address arsenic, a harmful heavy metal, to help solve wastewater issues. Other reported cases across Malaysia (2018-2024) are summarised in Table 1. The table demonstrates that the arsenic concentrations exceeded the permissible limit in most locations. Over the past six years (2018–2024), arsenic concentration grew increasingly serious. The data confirm that the previous conventional technologies used to eliminate arsenic were ineffective. Consequently, studying arsenic in Malaysia remains relevant, especially when it is now one of the most detected heavy metals in water bodies.

Table 1: Reported cases of arsenic contamination across Malaysia states from 2018 to 2024

Location	Arsenic Concentration (µg/L)	Reference
Klang River, Selangor	72	(Jesi et al., 2024)
Sepang River, Selangor	10.5	(Shukor, Krishnan, Shing, Ariffin, & Yong, 2023)
Langat River Basin, Selangor	34.5	(Ahmed et al., 2021)
Linggi River, Negeri Sembilan	1.31	(Razak, Aris, Zakaria, Wee, & Ismail, 2021)
Kuantan River, Pahang	42.30	(Yunus, Zuraidah, & John, 2020)
Tenom River, Johor	>100 ppm	(Y. Liang et al., 2020)
Kepayang River, Perak	0.288	(Affandi & Ishak, 2018)

2.1 Arsenic toxicity and human health effects

The toxicological effects of arsenic on human health are profound and multifaceted, ranging from acute symptoms, such as gastrointestinal distress, cardiovascular collapse, and neuropathies in cases of high-level exposure, to chronic conditions, including dermal lesions, respiratory diseases, diabetes, and particularly, various types of cancer resulting from long-term exposure to lower concentrations (Khan et al., 2022; L. Kumar et al., 2022; Mehdizadeh et al., 2022; Mojiri et al., 2024; Neumann et al., 2010). Arsenic acts by disrupting numerous cellular processes, including enzyme activity inhibition, oxidative stress induction, genotoxicity, and signal transduction interference, ultimately leading to the pathophysiological effects observed in exposed individuals. Furthermore, epidemiological studies have established strong correlations between long-term ingestion of arsenic-contaminated water and increased incidences of skin, lung, bladder, kidney, and liver cancers, emphasizing the urgent need for efficient and reliable removal technologies to safeguard public health (Sevak & Pushkar, 2024).

2.2 Regulations and guidelines for arsenic

In recognition of the severe health risks posed by arsenic, international regulatory bodies have imposed stringent standards to control its concentration in drinking water supplies. The World Health Organization (WHO) has established a guideline value of 10 µg/L for arsenic in drinking water, a standard that has been adopted by countries including Malaysia. The Malaysian Department of Environment mandates an effluent limit of 0.05 mg/L for arsenic discharge into upstream water bodies and 0.10 mg/L for downstream discharge (Ahmed et al., 2021). These limits are designed to minimise the risk of chronic arsenic toxicity while ensuring that treated water meets acceptable health and safety standards; nonetheless, consistently achieving such low concentrations remains a significant technical challenge for conventional water treatment systems (Ganie, Javaid, Hajam, & Reshi, 2023).

Hence, innovative water treatment methods such as PEUF that can selectively and efficiently remove arsenic ions even at very low concentrations are now highly desirable. Moreover, there is a need for sustainable solutions that minimise environmental impact and operational costs while ensuring compliance with evolving regulatory standards where arsenic contamination remains a persistent and escalating concern. This reinforces the development and optimisation of CMC-enhanced PEUF technologies as an area of critical research focus.

3 TECHNOLOGIES FOR ARSENIC REMOVAL

Over the past decades, a wide variety of technologies were developed and applied for the removal of arsenic from contaminated water systems. These technologies are based on different operational principles, such as chemical precipitation, adsorption, ion exchange, and membrane filtration. Nevertheless, these technologies often face significant challenges in achieving consistently high removal efficiencies under variable water quality conditions.

3.1 Coagulation-flocculation

Coagulation-flocculation processes, using chemicals, such as ferric chloride or alum, have been traditionally employed to remove arsenic by destabilising and aggregating arsenic-bearing particles into larger flocs that can be easily separated. Nonetheless, this method tends to be less effective for As(III) species unless a pre-oxidation step was introduced (Francisca & Carro Pérez, 2014; Yan et al., 2014). Additionally, coagulation processes generate considerable volumes of sludge requiring proper disposal, thus raising environmental and cost concerns associated with sludge management and secondary pollution risks (Skotta et al., 2023).

3.2 Adsorption

Adsorption technologies, employing media, such as activated alumina, iron hydroxides, and nanomaterials, have gained significant attention due to their high selectivity and operational simplicity. That said, the efficiency of adsorption processes is highly dependent on solution pH, the presence of competing ions, and the saturation of adsorbent surfaces (Bayuo, Rwiza, & Mtei, 2023; Kuriakose, Singh, & Pant, 2004; Szatyłowicz & Skoczko, 2017). While adsorption methods can achieve excellent arsenic removal rates at optimised conditions, issues related to media regeneration, and disposal of exhausted adsorbents continue to pose operational and environmental challenges (Yadav, Saidulu, Ghosal, Mukherjee, & Gupta, 2022). In addition to conventional adsorbents, recent research into novel materials, such as magnetic graphene oxide and iron-modified biochar composites has shown promise in enhancing adsorption capacities. Nevertheless, these materials often remain expensive or technically complex to produce on a large scale (Hira et al., 2023).

3.3 Oxidation-precipitation

Oxidation-precipitation processes represent another widely used technique, particularly for converting more toxic As(III) to the less toxic As(V) form, thereby improving subsequent removal efficiency by coagulation or adsorption. Nevertheless, this approach requires precise control of oxidation reactions and careful management of residual oxidants, such as chlorine or ozone (Bissen & Frimmel, 2003; Otgon, Zhang, & Yang, 2016; P. Singh, Zhang, Robins, & Hubbard, 2007). Furthermore, oxidation alone is rarely sufficient to meet drinking water standards and typically needs to be integrated with secondary treatment steps, which complicates the overall treatment train and increases both operational complexity and cost (Kurian, 2021; Mantzavinos & Psillakis, 2004; T. Wang et al., 2006). Similarly, ion exchange techniques using synthetic resins can selectively remove arsenic species; nevertheless, the performance is often hindered by competition from other anions (sulfate and phosphate) in natural waters, and resin fouling or breakthrough remains an operational concern (Ghurye &

Clifford, 2007; Staicu, Morin-Crini, & Crini, 2017). Table 2 presents a summarised comparison of major arsenic removal technologies, emphasizing their respective removal efficiencies, advantages, and limitations.

Table 2: Summary of arsenic removal technologies

Technology	Material	Advantages	Limitations
Coagulation-Flocculation	• Ferric Salts • Alum	• Effective at high concentration • Simple step water treatment	• Sludge production • Less Effective for As (III)
Adsorption	• Activated Alumina • Ferric Hydroxide	• High selectivity • Cost-effective	• Media regeneration needed • Sensitive to pH
Oxidation-precipitation	• Chlorine • Ozone	• Converts As (III) to As (V)	• Require additional treatment step
Membrane Filtration	• Polymeric Membrane	• High water quality • Easily achieve retention efficiency	• High cost • Membrane fouling
Polymer -Enhanced Ultrafiltration (PEUF)	• Synthetic Polymer • Biopolymer	• Selective removal • Polymer regeneration possible	• Polymer loss • Operational optimisation needed

3.4 Membrane filtration technologies

Membrane filtration technologies, including ultrafiltration (UF), nanofiltration (NF), and reverse osmosis (RO), have demonstrated excellent removal efficiencies for a wide range of contaminants, such as arsenic. Nevertheless, these technologies involve high operational costs, substantial energy consumption, and susceptibility to membrane fouling and scaling (Algieri et al., 2022; V. R. Moreira, Lebron, Santos, Coutinho de Paula, & Amaral, 2021; Zakhar, Derco, & Čacho, 2018). Among membrane-based processes, ultrafiltration alone is generally ineffective for removing ionic species due to the relatively large membrane pore size (Diallo, 2014; L. Liu et al., 2019). Nonetheless, by coupling ultrafiltration with a polymer complexation step, known as polymer-enhanced ultrafiltration (PEUF), it becomes possible to selectively remove target ions such as arsenic by increasing their apparent size through complex formation (Dilpazeer et al., 2023; Huang & Feng, 2019; Pookrod et al., 2005).

Although these processes have proven effective in removing arsenic from various wastewater sources, each innovation has several limitations, particularly with respect to metal ions removal efficiency (as detailed in Table 3) when using different membrane types and separation methods.

Table 3: Summary of methods used to remove arsenic from various membrane processes

Method	Water Source	Membrane	Initial Concentration of Arsenic (mg/L)	pH	Rejection %		Ref.
					As (III)	As (V)	
UF	Municipal Wastewater	PAC/FeCl3/UF	100	8	19	-	(Marjanović et al., 2023)
UF	Aqueous Solution	MEUF	20–50	7.6–7.9	89.32	-	(Yaqub & Lee, 2021)
UF	Drinking Water	Polyetherimide-based membrane	0.015–0.4	-	96.9	-	(Victor Rezende Moreira, Lebron, Santos, & Amaral, 2021)
MF	Contaminated Groundwater	PTFE membrane	20–50	7.6–7.9	96	-	(JT Wang, Kan, Lu, & Yang, 2019)
MF	Aqueous Solution	-	1.5	5	<90.7	-	(Pramod, Gandhimathi, Lavanya, Ramesh, & Nidheesh, 2020)
AGMD	Synthetic Industrial Wastewater	Air Gap membrane distillation (TF200, TF450 and TF1000)	2–100	5–9.5	96	-	(Alkhudhiri, Hakami, Zacharof, Abu Homod, & Alsadun, 2020)
NF	Synthetic Wastewater	Polyamide core-shell bio-functionalized mixed	-	9	99	-	(Zeeshan, Khan,

		membrane					Shafiq, & Sabir, 2020)
NF	Aqueous Solution	Membrane modified by graphene oxide/copper ferrodioxide	-	-	93.8	81.2	(Ghosh, Prabhakar, & Samadder, 2019)
NF	Sulphuric Acid Plant Wastewater	NF90	76	6	85	75	(Reis, Araújo, Vieira, Amaral, & Ferraz, 2019)
RO	Gold Mining Wastewater	TFC-HR and BW30	340	5	75 and 42	-	(Andrade et al., 2017)
RO	Mining Wastewater	DOW™ FILMTEC™ BW30-440i	-	-	96	-	(Samaei, Gato-Trinidad, & Altaee, 2020)
RO	Spent Geothermal Water	XLE BWRO	0.17	8	99	-	(Jarma et al., 2021)

4 POLYMER-ENHANCED ULTRAFILTRATION (PEUF)

4.1 Ultrafiltration process

Ultrafiltration (UF) system (Figure 2) is a membrane separation technique widely used in water and wastewater treatment processes, characterised by its ability to remove macromolecules, colloids, and particulates from aqueous solutions based on size exclusion of metal ions (Barjoveanu & Teodosiu, 2009; Ye et al., 2025; J. Zhang et al., 2023). UF membranes generally have pore sizes ranging from 1to 100 nm, enabling the retention of molecules with molecular weights greater than approximately 1,000 Daltons, while permitting smaller species, such as water and dissolved salts to pass through (Alebrahim & Moreau, 2023; Frenkel, 2010; Pérez-Gálvez, Guadix, Bergé, & Guadix, 2011). The driving force for the UF process is a relatively low applied pressure, typically ranging between 1to 10 bar, making it more energy-efficient than tighter membrane processes, such as nanofiltration or reverse osmosis (Cartwright, 2012; R. Wang, Cao, & Yan, 2019).

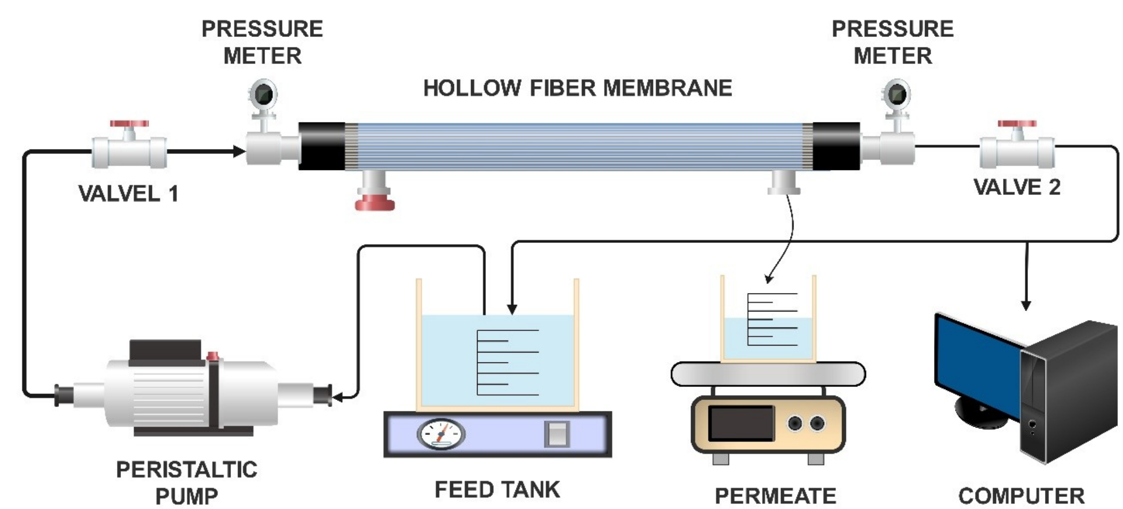

Figure 2: Schematic representation of the Ultrafiltration System

The membranes used for ultrafiltration are typically made from polymeric materials, such as polysulfone (PSf), polyvinylidene fluoride (PVDF), or cellulose acetate (CA), selected for their mechanical strength, chemical resistance, and antifouling properties. Nevertheless, membrane fouling remains an operational challenge that can compromise long-term process stability (Medina-Gonzalez, Aimar, Lahitte, & Remigy, 2011; Othman et al., 2020; Wu et al., 2018). However, when targeting small or neutrally charged species such as arsenite (As(III)) and arsenate (As(V)), UF alone becomes insufficient because these ions are smaller than the membrane pores and do not interact strongly enough with the membrane surface to be retained. In UF processes, a limitation arises when targeting small ionic contaminants, such as arsenic species, which are smaller than the membrane pores and thus cannot be rejected based on size exclusion alone. Therefore, to enable selective removal of arsenic from contaminated water, UF must be combined with a process that increases the apparent size of arsenic species, which is achieved by introducing a complexing polymer in the PEUF process.

4.2 Complexation-ultrafiltration process

The complexation–ultrafiltration process, forming the core principle of PEUF, involves the introduction of a water-soluble polymer into the contaminated feed solution in Figure 3. This polymer binds with target ions or molecules to form large macromolecular complexes that can then be retained by an ultrafiltration membrane (Huang & Feng, 2019; Korus, 2023, 2025; Li et al., 2021). These polymers typically contain functional groups, such as carboxyl, hydroxyl, sulfonate, or amine groups that are capable of chelating or electrostatically binding with metal ions, thereby increasing the effective size of the contaminant beyond the membrane's pore threshold (Korus, 2018; Korus, Dawczak, & Kępska, 2018; S. Kumar & Sharma, 2022; Rivas, Pereira, Cid, & Geckeler, 2005).

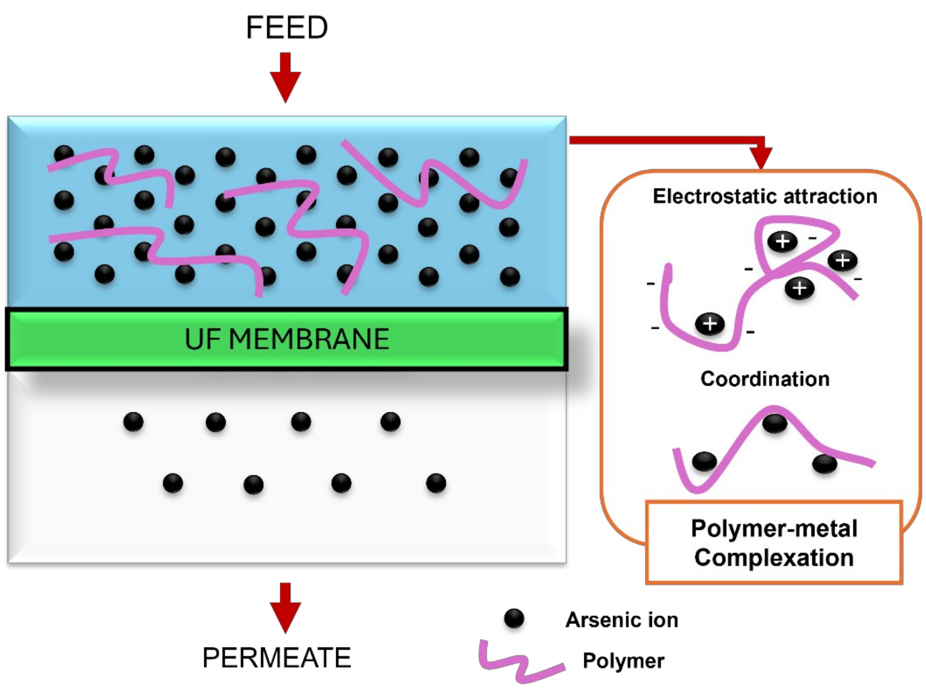

Figure 3: Complexation process of metal ions in the core principle of PEUF

Subsequent to complex formation, the feed mixture undergoes ultrafiltration, during which uncomplexed water and small neutral species permeate through the membrane, while the polymer-metal complexes are retained in the retentate stream (Huang & Feng, 2019; Korus, 2025). In the case of arsenic, the complexation step is particularly critical because arsenite and arsenate exist in different protonation states depending on pH. As(V), which exists primarily as negatively charged oxyanions ($H_2AsO_4^-$/$HAsO_4^{2-}$), readily interacts with positively charged or protonated polymer functional groups, while As(III), which exists mostly as neutral H_3AsO^3, forms weaker interactions and requires hydrogen bonding or coordination-type binding to achieve sufficient complex size for membrane retention.

Thus, complexation is not only a size-modifying step, but also the key determinant of selectivity, binding strength, and overall arsenic removal efficiency in PEUF systems. Factors influencing the efficiency of complexation include pH which affects ionization and polymer charge, ionic strength, polymer type and concentration, and the speciation and charge of target contaminants.

4.3 Selective polymer in PEUF process

Polymers utilised in PEUF must possess strong selectivity for the target contaminants to ensure efficient complex formation without excessive polymer dosage or loss. Ideal polymers exhibit high affinity binding towards specific metal ions while maintaining high solubility and stability under operational conditions (Huang & Feng, 2019; Korus & Piotrowski, 2017; Lam, Déon, Morin-Crini, Crini, & Fievet, 2018). A variety of polymers have been explored in the PEUF process, including polyethyleneimine (PEI), polyacrylic acid (PAA), chitosan, and carboxymethyl cellulose (CMC).

PEI, containing multiple amines groups, forms strong complexes with divalent and trivalent metal cations, whereas PAA, rich in carboxylate groups, binds well with cations, such as lead, cadmium, and arsenic species (Gao et al., 2022; Qin, Shao, He, & Li, 2012). A study by Korus (2023) emphasised the use of polyethyleneimine (PEI) and evaluated its performance for the removal of Cu(II), Cd (II), Zn (II), Ni(II) and Cr(II) in PEUF. It demonstrated superior selectivity and removal efficiency compared to conventional polymers with 67% to 91% of concentrated metal ions, attributing this to the polymer's unique functional groups (EDTA, SO_4^{2-}, PO_4^{3-}) which facilitated stronger and more specific interactions with the metal ions.

Furthermore, natural biopolymers, such as chitosan and CMC, offer additional advantages due to their biodegradability and low toxicity. Recent studies by Al-Mutair, Kumar, Al-Mur, Zubair, and Barakat (2024) demonstrated the use of unmodified CMC with Iron (Fe) have higher removal efficiency at 99.05% under optimised conditions at pH 7. However, the existing discussion in previous studies has tended to focus more on removal of heavy metals rather than arsenic, which creates a gap that must be addressed. Therefore, in the context of arsenic removal, CMC is especially relevant because its carboxylate ($-COO^-$) groups can interact with arsenate ions via electrostatic and coordination interactions, while its hydroxyl groups can form hydrogen bonds with arsenite, enabling selectivity based on arsenic oxidation state.

This selectivity becomes particularly meaningful in PEUF because As(V) forms stronger and more stable complexes with CMC than As(III), explaining As(V) is generally removed at higher efficiency unless pre-oxidation is performed. Tables 4 summarises various studies on heavy metal removal using different polymers in PEUF under diverse experimental conditions which include the type of polymer used, metal ion species, initial metal ion concentrations, polymer dosage, membrane material and molecular weight cut-off (MWCO).

Table 4: Selective polymers for complexation in PEUF process

Polymer	Origin	Metal	[Metal][a]	[Polymer][b]	Membrane	MWCO[c]	pH	Removal %	References
CMC	Synthetic	Ni^{2+}	1.7×10^{-5} mol/L	2×10^{-1} mol/L	PA-TFC	3.5	7.4	≥69	(Lam et al., 2018)
CMC	Synthetic	Fe	133.15 mg/L	0.455 g/L	PVDF	-	4–7	99.05	(Al-Mutair et al., 2024)
CMC	Synthetic	Cr^{3+}	30 mg/L	2 mg/L	PES	10	≥7	99.5	(Xanthopoulou & Katsoyiannis, 2019)
CMC	Synthetic	Ni^{2+} Cu^{2+} Cr^{3+}	25 mg/L	1 g/L	PES	10	7	94.4 98 98.3	(Barakat & Schmidt, 2010)
CMC	Synthetic	Cu^{2+}	200 mg/L	1200 mg/L	PAN	10	2	99.2	(Petrov & Nenov, 2004)
CMC	Synthetic	Fe^{2+}	20 mg/L	0.5 %	PDVF	150	4.8	80	(Konovalova, Kolesnyk, Ivanenko, & Burban, 2018)
Chitosan	Synthetic	Cr^{6+}	1 mg/L	278 mg/L	PES	5	5.9	100	(Köker & Cebeci, 2023)
Chitosan	Synthetic	Cu^{2+} Ni^{2+}	32 mg/L 29 mg/L	1 g/L	PEI	3 10	5–8	60 40	(Zamariotto, Lakard, Fievet, & Fatin-Rouge, 2010)
Chitosan	Synthetic	Cu^{2+} Ni^{2+} Cr^{6+}	41 mg/L 35 mg/L 43 mg/L	123 mg/L 105 mg/L 258 mg/L	PES	50	10 10 4	100 99 95	(Kavitha, Dalmia, Samuel, Prabhakar, & Rajesh, 2020)

Chitosan	France Chitine	Ni^{2+}	1.7×10^{-5} mol/L	2×10^{-1} mol/L	PA-TFC	3.5	5.4	69	(Lam et al., 2018)
Chitosan	Synthetic	Zn^{2+} Pb^{2+}	70 mg/L	35 mg/L	PES	50	2	95 98	(Kavitha, Kedia, Babaria, Prabhakar, & Rajesh, 2020)
PEI	Synthetic	Ni^{2+}	11.5 mg/L	57.5 mg/L	PES	30	8	93	(Shao et al., 2013)
PEI	Synthetic	Cr^{3+}	10.4 mg/L	1.49 mg/L	Cell Membrane	-	2	91.65	(Korus, 2023)
PEI	Synthetic	Co^{2+}	65 mg/L	4 mg/L	RC	5	6.4	99.65	(Cojocaru, Zakrzewska-Trznadel, & Jaworska, 2009)
PEI	Synthetic	NO_3^-	20 mg/L	0.1 wt%	PSf	6	4	92.3	(Li et al., 2021)
PAA	Synthetic	Cu^{2+} Cd^{2+} Zn^{2+}	1×10^{-3} mol/L	2×10^{-3} mol/L	PES	10	5	80 93 70	(Ennigrou, Sik Ali, Dhahbi, & Ferid, 2015)
PAA	Synthetic	Cu	100 mg/L	50 mg/L	PSf	5	6.9	90	(Mamaghani, Bidhendi, & Mehrdadi, 2023)
PA	Synthetic	Co^{2+}	10 mg/L	1 mg/L	PES	10	6	95	(Dambies, Jaworska, Zakrzewska-Trznadel, & Sartowska, 2010)

[a]Initial metal ion concentrations, [b]Polymer concentrations, [c]Molecular weight cut-off (kDa)

4.4 Advantage and disadvantage of PEUF

Despite its numerous advantages, PEUF presents certain challenges that need to be considered during system design and operation. The advantages and disadvantages of the PEUF process are summarised in Table 5. As a result, PEUF has gained significant attention for addressing diverse environmental and industrial challenges, including water purification and wastewater treatment, while contributing to the advancement of membrane technologies.

Table 5: Selective polymers for complexation in PEUF process

Advantages	Disadvantages
• High selectivity towards targeted metal ions, depending on polymer type • Lower pressure and energy demand compared to Reverse Osmosis (RO) • Potential for continuous polymer reuse, which can reduce excessive sludge production • Moderate compared to multi-step conventional water treatment that can reuse the polymer • Single-stage preparation that requires no extensive chemical addition	• Careful optimisation of polymer type and concentration is required • Operational complexity can be introduced by the polymer recovery process • Risk of polymer loss to the permeate if complexation is incomplete • High initial costs for high-quality membranes and specialised polymers • Ongoing challenges with membrane fouling and maintenance.

5 CARBOXYMETHYL CELLULOSE (CMC) IN PEUF

5.1 Why use CMC as biosorbent?

Carboxymethyl cellulose (CMC) has demonstrated high effectiveness as a polymeric ligand in PEUF processes, offering a range of advantageous properties that make it particularly suitable for the selective removal of metal ions, such as arsenic from aqueous environments (Kurhade, Shaikh, Nagulwar, & Kale, 2024). Derived from natural cellulose, CMC (Figure 4) is an anionic, water-soluble polysaccharide characterised by the presence of carboxymethyl groups attached to its glucose units. These functional groups impart high hydrophilicity and a strong negative charge density over a broad pH range (Mehrabi et al., 2024; Revathi et al., 2024). These functional groups serve as active binding sites for positively charged or even negatively charged contaminants under specific conditions, enabling the formation of stable, high-molecular-weight complexes that can be effectively retained by ultrafiltration membranes during the PEUF process. In this study, CMC does not form strong covalent bonds because both arsenate and CMC carry negative charges. Hence, arsenic can interact with CMC through weaker forces such as hydrogen bonding and van der Waals interactions, and with the help of multivalent cations that act as bridges. These interactions allow arsenic to associate with the polymer, making it large enough to be retained by the ultrafiltration membrane.

However, previous studies have shown that the nature and strength of interaction between CMC and arsenic depend strongly on arsenic speciation, where As(V) species ($H_2AsO_4^-$/$HAsO_4^{2-}$) form more stable complexes with protonated hydroxyl groups and localised hydrogen-bonding networks, while As(III), existing mostly as neutral H_3AsO_3 at near-neutral pH, interacts through weaker dipole and hydrogen bonding forces. Therefore, while CMC does not "chelate" arsenic in the same manner as metal cations, the cumulative effect of hydrogen bonding, electrostatic interactions in localized protonated regions, and cation bridging has been experimentally demonstrated to produce sufficiently stable macromolecular CMC–As complexes capable of membrane rejection.

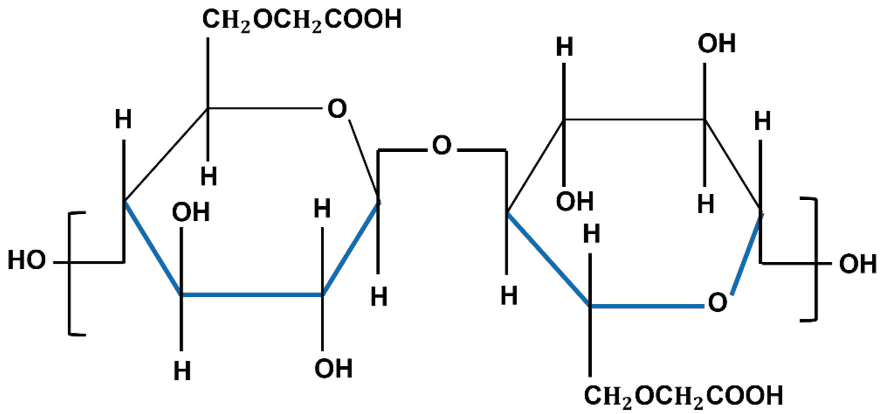

Figure 4: The molecular structure of Carboxymethyl cellulose (CMC).

Carboxymethyl cellulose (CMC) is particularly attractive as a polymer in the PEUF process due to its unique combination of environmental friendliness, high functional group density, excellent binding capacity, and cost-effectiveness (Ab Rasid et al., 2021; Kurhade et al., 2024). As a derivative of natural cellulose, CMC is an anionic polyelectrolyte with its abundant carboxymethyl groups exhibit strong interactions with a wide range of metal ions, making it highly effective in complexing both free metal cations and certain anionic species under specific pH conditions (Al-Mutair et al., 2025; Iurciuc, Atanase, & Popa, 2019). Unlike synthetic polymers, such as PEI or PAA, CMC offers significant advantages in terms of biodegradability and non-toxicity, aligning well with modern green chemistry approaches aimed at minimising secondary environmental pollution. Moreover, the water high solubility of CMC across a broad pH range ensures operational flexibility, while its low cost and wide availability make it especially attractive for large-scale water treatment applications, particularly in developing regions (Ahmad, Singh, Manral, Pandey, & Rashid, 2023; Rahman et al., 2021; Tyagi & Thakur, 2023).

In the study of arsenic removal, the biosorbent of CMC is particularly advantageous because arsenic-containing sludge produced from coagulation or precipitation techniques is often classified as hazardous waste, whereas PEUF using CMC minimizes secondary waste generation and allows the polymer solution to be regenerated and reused through controlled pH adjustment. As reported by Mendes et al. (Mendes, Caldeira, & Ciminelli, 2021), industrial wastewater treatment using ferric based coagulation can generate substantial amount of Fe-As sludge, typically ranging between 10-20 g/L, which requires subsequent dewatering, chemical stabilization, and controlled disposal to prevent the re-release of arsenic into downstream effluents. In contrast, PEUF using CMC minimizes secondary waste generation because the polymer-arsenic complexes remain fully in the aqueous phase and can be regenerated through controlled pH adjustment, allowing the polymer solution to be reused without producing large quantities of excess sludge in PEUF system.

5.2 Free metal cations and anionic species

In PEUF processes employing CMC, complexation can occur with a wide range of metal ions beyond arsenic, including lead (Pb^{2+}), cadmium (Cd^{2+}), arsenate (As^{3+}), and chromium species (Cr^{3+}/CrO_4^{2-}). Studies have shown that CMC binds efficiently with free cations through the formation of coordination bonds between the carboxyl groups on the polymer backbone and the metal ions, creating stable metal-polymer complexes resistant to dissociation under ultrafiltration conditions (Khvan, Madzhidova, & Turaev, 2005; K. Singh, Kumar, Mishra, & Gupta, 2020). In the case of anionic species, such as arsenate ($H_2AsO_4^-$) or chromate (CrO_4^{2-}), the binding mechanism involves hydrogen bonding or electrostatic interactions, depending on the ionization state of the CMC polymer at different pH levels.

For arsenic specifically, the removal efficiency is strongly dependent on pH because protonation and deprotonation influences both arsenic speciation and the ionization state of the CMC carboxyl groups. As(V) shows increased affinity under slightly acidic to neutral conditions, where partial protonation allows hydrogen bonding, whereas As(III) removal is enhanced under neutral conditions where hydrogen bonding predominates. In this study, As(III) does not carry a charge and therefore cannot interact with CMC through ionic attraction; instead, its association with CMC occurs mainly through hydrogen bonds formed between the hydroxyl groups of CMC and the hydroxyl groups of H_3AsO_3. These hydrogen-bonding interactions allow As(III) to associate with the polymer matrix long enough to form larger macromolecular complexes that can be retained by the UF membrane in the PEUF system.

Hence, studies have demonstrated that CMC's anionic nature favors the adsorption or complexation of cationic species more strongly. Nonetheless, by carefully adjusting pH conditions, effective removal of anionic arsenic species can be achieved (Rahman et al., 2021; H. Zhang et al., 2024). The versatility of CMC in interacting with both types of species broadens its applicability across a broad spectrum of heavy metal contaminations commonly found in real-world wastewaters. Recent advancements have shown that through targeted polymer modifications, the adsorption efficiency of CMC can be substantially enhanced, further expanding its potential for various applications (Akter et al., 2021).

However, it is important to clarify that the removal of arsenic using CMC is not solely governed by simple electrostatic attraction. Meanwhile, the mechanism is multipoint and cooperative, involving hydrogen bonding, cation bridging such as Ca^{2+} and Fe^{3+}, localized electrostatic interactions, and network adsorption, which describes the formation of a loose three-dimensional, entangled CMC network in solution whose overlapping polymer segments create microdomains that physically trap arsenic species via simultaneous weak interactions at many binding sites along the polymer backbone. These microdomains enable neutral As(III) (H_3AsO_3)to form multisite hydrogen bonds with $-OH$ and $-COOH$ groups in CMC, linking a single As(III) molecule to multiple polymer chains and producing macromolecular complexes that are retained by UF membranes.

5.3 Metal-polymer complexes and mechanism

The mechanism underlying metal binding by CMC involves multiple interactions, primarily coordination bonding between the oxygen atoms of the carboxylate groups and metal cations, supplemented by electrostatic attraction and hydrogen bonding, particularly in the case of anionic contaminants (S. Liang et al., 2024; Y. Liu, Wang, & Wang, 2010; Jian Wang, Liu, Duan, Sun, & Xu, 2019). In the study of As(III), its binding behaviour is distinct from that of conventional cationic metals. At near-neutral pH, As(III) exists predominantly as arsenous acid (H_3AsO_3) which is a neutral, trivalent hydroxyl-bearing molecule rather than a free As^{3+} cation. As a result, As(III) does not undergo electrostatic attraction or classical ion-exchange with the carboxylate groups ($-COO^-$) of CMC. Thus, the dominant mechanism involves the formation of multisite hydrogen bonding interactions between the hydroxyl groups of H_3AsO_3 and the hydroxyl and carboxyl functional groups of CMC by initiating a reversible association (Equation 1). Because H_3AsO_3 contains three hydroxyl groups, it can interact simultaneously with multiple CMC sites. These cooperative hydrogen-bond interactions allow a single As(III) molecule to serve as a linking node between different CMC chains, leading to the formation of macromolecular polymer–arsenite complexes in Equation 2.

$$CMC-OH+\ H_2AsO_3 \rightleftharpoons CMC-OH\ \cdots As\left(OH\right)_2OH \tag{1}$$

$$n\left(CMC-OH\right)+\ H_2AsO_3 \rightleftharpoons \left[CMC-OH\ \cdots As\left(OH\right)_2\cdots HO-CMC\right]n \tag{2}$$

$$CMC-COO^{-}+\ H_2AsO_3 \rightleftharpoons CMC-COO\ \ \rightarrow As\left(OH\right)_2+OH^{-} \tag{3}$$

The formation of these extended hydrogen-bonded networks results in an increase in the hydrodynamic radius of the polymer–arsenite complexes. This enlarged macromolecular complex is retained by the ultrafiltration membrane even though the free As(III) species is small enough to permeate. Thus, the key factor enabling As(III) removal in PEUF is structural association leading to size enlargement, rather than charge-driven interactions. In localized regions where carboxylate groups on CMC are deprotonated, inner-sphere ligand exchange may also occur, forming a weak coordination bond in which one hydroxyl group from H_3AsO_3 is replaced in Equation 3.

Although this coordination pathway is weaker than metal–carboxylate complexation observed with divalent metal ions such as Cu^{2+}, it contributes additional stability to the macromolecular CMC-As(III) complex. The extent of this interaction is influenced by pH, where mild deprotonation of carboxyl groups promotes ligand exchange while still preserving the hydrogen-bond network that drives complex formation.

Additionally, CMC demonstrates good reusability, with only minimal loss in rejection capacity over multiple filtration cycles, thereby reducing sludge management. The interaction mechanisms are not limited to electrostatic attraction; hydrogen bonding and coordination bonding may also occur, particularly in cases involving multivalent metals or partially protonated functional groups. Metal precipitation can also be observed on the membrane surface post-filtration, further contributing to contaminant rejection.

CMC's compatibility with ultrafiltration membranes is enhanced when blended with other polymers, improving membrane hydrophilicity, water flux, and antifouling properties. These synergistic effects not only support high selectivity and recovery, thereby facilitating broader applicability in real wastewater treatment. The use of CMC in PEUF offers numerous advantages over conventional techniques due to its high binding efficiency, low energy requirements, environmentally friendly nature, and economic practicality. Future studies should therefore prioritise detailed mechanistic evaluation of CMC-As interactions, including the identification of specific functional groups involved in binding, quantification of interaction energies through spectroscopic shifts by using FTIR or XPS, and clarification of whether As(III) forms monodentate, bidentate, or multi-site coordination with CMC chains. This deeper mechanistic insight is essential for optimising operational pH adjustment for selective As(III) removal and for guiding pilot-scale implementation to assess long-term membrane performance and polymer regeneration feasibility.

6 CHALLENGES AND FUTURE DIRECTIONS IN PEUF

Despite the promising capabilities of CMC-PEUF systems in removing arsenic from aqueous environments, several critical challenges hinder their widespread implementation. One of the foremost issues is membrane fouling, which occurs due to the accumulation of metal-polymer complexes on the membrane surface, leading to reduced permeate flux and increased operational maintenance. Furthermore, the loss of unbound CMC during filtration and the difficulty in its recovery reduce the economic and environmental viability of the process. The selectivity of CMC for arsenic, particularly in the form of arsenate, may also decline in complex wastewater matrices where competing ions are present, disrupting the efficiency of metal binding. Additionally, real wastewater streams often contain co-existing oxyanions such as phosphate and sulfate, which compete with arsenate for interaction sites on CMC, thereby lowering selectivity and requiring precise pH and dosage control to sustain high removal efficiency in field applications.

Future research should focus on enhancing the chemical structure of CMC through functional group modifications, such as amination or thiolation, that can improve its affinity for arsenic species under a wider range of pH and ionic strengths. Therefore, functional modification strategies have shown potential in strengthening both coordination and hydrogen-bonding interactions, particularly with As(III), thereby addressing the key mechanistic limitation of weak and non-specific binding between unmodified CMC and neutral As(III) species which reduces complex stability and leads to lower removal efficiency in the CMC-PEUF system. The development of hybrid PEUF systems that integrate advanced oxidation processes (AOPs), adsorption units, or electrochemical methods could address multiple water quality parameters simultaneously, improving treatment robustness. In particular, coupling PEUF with AOPs with UV/H_2O_2 and MnO_2 oxidation to pre-oxidize As(III) to As(V) can significantly enhance selectivity, since As(V) forms more stable complexes with CMC, as demonstrated in integrated AOP–membrane systems. Additionally, designing systems with reusable or regenerable membrane-polymer configurations would enhance sustainability and lower treatment costs. Long-term reuse of CMC requires stable polymer recovery steps, such as pH-induced desorption cycles or diafiltration loops, which must be optimized to avoid polymer degradation or loss.

Economic and scalability considerations are also necessary to support real-world deployment. While PEUF operates at lower pressure and energy consumption than reverse osmosis, polymer consumption and membrane cleaning cycles contribute significantly to operational costs. Cost models and life-cycle assessments are therefore needed to quantify polymer recovery efficiency, fouling management costs, and overall treatment economics in comparison with coagulation–filtration and adsorption systems. The feasibility of scaling up CMC-PEUF for municipal and industrial applications must be evaluated through life-cycle assessments and cost-benefit analyses. Pilot-scale demonstrations using real water matrices, continuous-flow UF modules, and multi-cycle regeneration trials are essential to validate long-term membrane stability, polymer lifetime, and arsenic rejection performance. Moreover, investing in the development of optimised CMC-based PEUF systems offers a path toward a more accessible, sustainable, and effective solution for global arsenic contamination challenges.

7 CONCLUSIONS

This review has demonstrated that polymer-enhanced ultrafiltration using carboxymethyl cellulose (CMC) represents a sustainable and effective method for the removal of arsenic from aqueous environments. The unique functional groups present in CMC enable interactions with arsenic species through hydrogen bonding, localized electrostatic attraction, and cation-bridging, forming macromolecular complexes that can be retained by ultrafiltration membranes. The reported removal efficiencies of CMC-PEUF systems, reaching 94-98% for As(V) and 82-99% for As(III), highlight the strong potential of this method in meeting regulatory discharge standards when operating conditions are carefully optimized. However, challenges related to polymer loss, membrane fouling, and reduced selectivity in the presence of competing ions must be addressed before large-scale implementation can be achieved.

Future advancements in CMC modification, such as introducing thiol or amine functional groups, may strengthen interactions with arsenic species and expand applicability across a wider pH and ionic strength range. Additionally, the integration of PEUF with advanced oxidation processes (AOPs), particularly those capable of converting As(III) to As(V), and hybrid membrane systems could significantly enhance treatment robustness and selectivity. Economic feasibility and scalability should be supported through pilot-scale evaluations, including small-scale hollow fiber membrane in PEUF systems that operated under continuous flow to monitor CMC carry-over and long-term rejection stability, life-cycle assessments that compare chemical consumption and sludge generation of CMC-PEUF against conventional coagulation methods treating arsenic-contaminated industrial effluent, cost-benefit analyses that quantify operational costs associated with CMC regeneration cycles, and energy demand of low-pressure UF operation. These assessments also require investigation of membrane module over repeated CMC-arsenic filtration cycles, such as monitoring flux decline, recovery ratio, and fouling reversibility, to determine whether the system can maintain consistent As(III) removal efficiency under realistic wastewater conditions.

Lastly, CMC-PEUF shows strong promise as a low-energy, environmentally friendly, and adaptable treatment strategy for arsenic-contaminated water. Continued research focused on process optimization, such as refining polymer dosage, membrane selection, and operating pH to maximise arsenic-CMC complex formation and system integration, such as combining PEUF with oxidation units or pre-filtration steps to stabilise long-term performance which will be essential to translate this approach into practical in full-scale wastewater treatment applications where the system must operate continuously at higher flow rates, handle variable wastewater compositions, and maintain consistent arsenic removal efficiency under real operational in wastewater treatment.

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COMPETING INTERESTS

Author declares that the have no competing interests that are relevant to the content of this chapter. The corresponding author, Nurul Huda Baharuddin have confirmed on behalf of all authors that are no financial or personal relationship that could have influenced the work reported in this manuscript.

ETHICAL APPROVAL

This manuscript is a review article and does not involve human participants, animals or any primary experimental data requiring ethical approval. Therefore, ethics approval, consent to participate and consent to publish are not applicable. The corresponding author, Nurul Huda Baharuddin have confirmed that no ethical approval was required, and no participant data were collected, used or published in this study.

DATA AVAILABILITY

Not applicable. This manuscript is a review article that systematically analyses and synthesises findings from previously published and publicly accessible scientific literature.

No new experimental work, field sampling, or numerical datasets were generated or analysed as part of this study. All data discussed are derived from the original sources cited throughout the manuscript.

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b. Author Contributions

Nor Husnina Nor Shamsi was responsible for the overall conceptualization of the review, conducting the literature analysis, writing the full manuscript text, and preparing all figures and tables. Nurul Huda Baharuddin contributed by providing supervision, critical manuscript revision, and editorial guidance throughout the preparation of the work. Zainura Zainon Noor, Norzita Ngadi, and Mohd Yusof Baharuddin supported the study by reviewing the manuscript and offering constructive feedback to improve clarity and academic quality. All authors read and approved the final version of the manuscript.

c. Consent to Participate

Not applicable. This manuscript is a review article based exclusively on the analysis, synthesis, and interpretation of previously published scientific literature. It does not involve the recruitment of human participants, the use of human or animal subjects, surveys, interviews, experiments, or the collection of primary data that would require informed consent.

d. Consent to Publish

Not applicable. This manuscript does not include any personal, clinical, or identifiable information relating to individual persons. All information discussed is derived from publicly available, peer-reviewed sources, and therefore no consent for publication is required.

NOMENCLATURE

AOP	-	Advanced Oxidation Process
As(III)	-	Arsenite
As(V)	-	Arsenate
As	-	Arsenic
CA	-	Cellulose Acetate
CMC	-	Carboxymethyl Cellulose
DOE	-	Department of Environmental Malaysia
H_3AsO_3	-	Arsenous acid
$H_2AsO_4^-/HAsO4^{2-}$	-	Arsenate oxyanions
NF	-	Nanofiltration
PAN	-	Polyacrylonitrile
PAA	-	Polyacrylic Acid
PA-TFC	-	Polyamide Thin-Film Composite Membrane
PEI	-	Polyethyleneimine
PEUF	-	Polymer-Enhanced Ultrafiltration
PES	-	Polyethersulfone

PDADMAC	-	Poly(dimethyldiallylammonium chloride)
PSf	-	Polysulfone
PVDF	-	Polyvinylidene Fluoride
RO	-	Reverse Osmosis
SEM	-	Scanning Electron Microscopy
UF	-	Ultrafiltration
WHO	-	World Health Organization

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